```
;-*- Mode: LISP; Syntax: ANSI-Common-Lisp; Package: CL-USER; Base: 10 -*-

Priorities

  [0] Not associated with a priority.
  [1] Critical.  Very visible.        Must fix                 Show stopper.
  [2] Important. Relatively Visible.  Should fix.              Document for first release.
  [3] Medium.    Visible.             Fix sometime.            Document until fixed.
  [4] Low.       Rare or unimportant. Fix when resources.      Document until fixed.
  [5] Foo.       As good as we are likely to get in Genera.    Document forever.
  [6] Non-bug.   Optional feature.    Should implement.        Document until implemented.

Believed to be generally conforming

 DOCUMENTATION VARIABLE TYPE

 STREAM-EXTERNAL-FORMAT

 EVAL
 COMPILE
 WITH-COMPILATION-UNIT

 LOAD-TIME-VALUE

 CONSTANTP

 MACRO-FUNCTION

 DEFINE-COMPILER-MACRO
 COMPILER-MACRO
 COMPILER-MACRO-FUNCTION

 DEFINE-MODIFY-MACRO

 &AUX
 &BODY
 &ENVIRONMENT
 &WHOLE
 LAMBDA-LIST-KEYWORDS

 * ** *** (variables)
 + ++ +++ (variables)
 / // /// (variables)
 - (variable)

 + - * / (functions)
 1+ 1-

 DOTIMES DOLIST

 DEFTYPE

 CASE OTHERWISE TYPECASE

 MINUSP ZEROP PLUSP
 EVENP ODDP

 *BREAK-ON-SIGNALS*

 *LOAD-PATHNAME*  *COMPILE-FILE-PATHNAME*
 *LOAD-TRUENAME*  *COMPILE-FILE-TRUENAME*
 *LOAD-VERBOSE*   *COMPILE-VERBOSE*

 *DEBUGGER-HOOK*
 *PACKAGE*

 WITH-STANDARD-IO-SYNTAX

 *PRINT-ESCAPE*    *PRINT-BASE*   *PRINT-LENGTH*  *PRINT-ARRAY*   *PRINT-CIRCLE*  *PRINT-CASE*
 *PRINT-READABLY*  *PRINT-RADIX*  *PRINT-LEVEL*   *PRINT-GENSYM*  *PRINT-PRETTY*

 BYTE
 BYTE-POSITION BYTE-SIZE
 DEPOSIT-FIELD  MASK-FIELD
 DPB LDB LDB-TEST
```

```
DESTRUCTURING-BIND

DEFINE-SETF-EXPANDER
GET-SETF-EXPANSION

*READ-EVAL*
*READ-SUPPRESS*
*READ-DEFAULT-FLOAT-FORMAT*
*READTABLE*

TYPEP    SUBTYPEP
TYPE-OF
SATISFIES MEMBER OR AND EQL NOT (type specifiers)
FTYPE

DEFSTRUCT

GENSYM  *GENSYM-COUNTER*
GENTEMP

CAR  CAAR  CAAAR  CAAAAR  CDAAAR
CDR  CADR  CAADR  CAAADR  CDAADR
     CDAR  CADAR  CAADAR  CDADAR
     CDDR  CADDR  CAADDR  CDADDR
           CDAAR  CADAAR  CDDAAR
           CDADR  CADADR  CDDADR
           CDDAR  CADDAR  CDDDAR
           CDDDR  CADDDR  CDDDDR

GET-INTERNAL-REAL-TIME
GET-INTERNAL-RUN-TIME
INTERNAL-TIME-UNITS-PER-SECOND

FIRST SECOND THIRD FOURTH FIFTH SIXTH SEVENTH EIGHTH NINTH TENTH
REST

CONS LIST* RPLACA RPLACD
CONSP ATOM
COPY-TREE  TREE-EQUAL

LIST LISTP
MAKE-LIST  COPY-LIST
MEMBER MEMBER-IF MEMBER-IF-NOT
LAST LDIFF
BUTLAST NBUTLAST
TAILP
NTH NTHCDR
APPEND NCONC
REVAPPEND NRECONC
LIST-LENGTH

LENGTH
REVERSE NREVERSE
SEARCH
REPLACE
FILL

SORT STABLE-SORT

MACHINE-INSTANCE
LONG-SITE-NAME   MACHINE-TYPE    SOFTWARE-TYPE     LISP-IMPLEMENTATION-TYPE
SHORT-SITE-NAME  MACHINE-VERSION SOFTWARE-VERSION  LISP-IMPLEMENTATION-VERSION

PROBE-FILE  DELETE-FILE  RENAME-FILE
DIRECTORY
TRUENAME
FILE-AUTHOR
FILE-LENGTH
FILE-POSITION
FILE-WRITE-DATE

COMPLEMENT CONSTANTLY

ACONS PAIRLIS COPY-ALIST
 ASSOC  ASSOC-IF  ASSOC-IF-NOT
RASSOC RASSOC-IF RASSOC-IF-NOT
```

```
PATHNAME  PATHNAMEP  MAKE-PATHNAME
PATHNAME-HOST  PATHNAME-DEVICE  PATHNAME-DIRECTORY
PATHNAME-NAME  PATHNAME-TYPE    PATHNAME-VERSION
PATHNAME-MATCH-P
WILD-PATHNAME-P
NAMESTRING        DIRECTORY-NAMESTRING  HOST-NAMESTRING
PARSE-NAMESTRING  FILE-NAMESTRING       ENOUGH-NAMESTRING
USER-HOMEDIR-PATHNAME
MERGE-PATHNAMES

DECODE-UNIVERSAL-TIME
ENCODE-UNIVERSAL-TIME
GET-UNIVERSAL-TIME
GET-DECODED-TIME

READTABLE READTABLEP
COPY-READTABLE

GET-MACRO-CHARACTER   SET-DISPATCH-MACRO-CHARACTER   MAKE-DISPATCH-MACRO-CHARACTER
SET-MACRO-CHARACTER   GET-DISPATCH-MACRO-CHARACTER   SET-SYNTAX-FROM-CHAR

GET   REMPROP  GET-PROPERTIES
GETF  REMF

FLET LABELS

VECTOR VECTORP
VECTOR-PUSH  VECTOR-POP  VECTOR-PUSH-EXTEND

LEAST-NEGATIVE-DOUBLE-FLOAT             LEAST-POSITIVE-DOUBLE-FLOAT
LEAST-NEGATIVE-LONG-FLOAT               LEAST-POSITIVE-LONG-FLOAT
LEAST-NEGATIVE-SHORT-FLOAT              LEAST-POSITIVE-SHORT-FLOAT
LEAST-NEGATIVE-SINGLE-FLOAT             LEAST-POSITIVE-SINGLE-FLOAT

LEAST-NEGATIVE-NORMALIZED-DOUBLE-FLOAT  LEAST-POSITIVE-NORMALIZED-DOUBLE-FLOAT
LEAST-NEGATIVE-NORMALIZED-LONG-FLOAT    LEAST-POSITIVE-NORMALIZED-LONG-FLOAT
LEAST-NEGATIVE-NORMALIZED-SHORT-FLOAT   LEAST-POSITIVE-NORMALIZED-SHORT-FLOAT
LEAST-NEGATIVE-NORMALIZED-SINGLE-FLOAT  LEAST-POSITIVE-NORMALIZED-SINGLE-FLOAT

SPECIAL-OPERATOR-P

APROPOS APROPOS-LIST
BREAK
DISASSEMBLE
DRIBBLE
ED
FIND-ALL-SYMBOLS
INSPECT
ROOM
STEP
TIME
TRACE UNTRACE

RANDOM-STATE MAKE-RANDOM-STATE RANDOM-STATE-P
RANDOM *RANDOM-STATE*

ABS SIGNUM
MOD (function)  REM
SQRT ISQRT
ASH
COS COSH  ACOS ACOSH
SIN SINH  ASIN ASINH
TAN TANH  ATAN ATANH
LOG EXP   EXPT
PI
PHASE
CONJUGATE
CIS
GCD LCM

HASH-TABLE HASH-TABLE-P
HASH-TABLE-COUNT  HASH-TABLE-REHASH-SIZE       HASH-TABLE-TEST
HASH-TABLE-SIZE   HASH-TABLE-REHASH-THRESHOLD
GETHASH           REMHASH
MAPHASH           WITH-HASH-TABLE-ITERATOR     CLRHASH
```

```
SXHASH

BOOLE
BOOLE-1 BOOLE-AND BOOLE-NAND BOOLE-ANDC1 BOOLE-C1 BOOLE-CLR BOOLE-ORC1 BOOLE-EQV
BOOLE-2 BOOLE-IOR BOOLE-NOR  BOOLE-ANDC2 BOOLE-C2 BOOLE-SET BOOLE-ORC2 BOOLE-XOR

*TERMINAL-IO*

T NIL
AND OR NOT

IF COND WHEN UNLESS

EVERY NOTEVERY
SOME  NOTANY

PROCLAIM DECLAIM
THE LOCALLY

DYNAMIC-EXTENT

UNWIND-PROTECT

SLEEP

CATCH THROW
BLOCK RETURN-FROM
TAGBODY GO

EQ EQL

NULL ENDP

MAP-INTO
MAPC MAPCAR  MAPCAN
MAPL MAPLIST MAPCON

LOOP LOOP-FINISH

ARRAY-DIMENSION-LIMIT   CALL-ARGUMENTS-LIMIT     MULTIPLE-VALUES-LIMIT
ARRAY-RANK-LIMIT        CHAR-CODE-LIMIT
ARRAY-TOTAL-SIZE-LIMIT  LAMBDA-PARAMETERS-LIMIT
UPGRADED-ARRAY-ELEMENT-TYPE

SYMBOL SYMBOLP
SYMBOL-FUNCTION  SYMBOL-NAME     SYMBOL-PLIST
SYMBOL-VALUE     SYMBOL-PACKAGE
 BOUNDP MAKUNBOUND
FBOUNDP FMAKUNBOUND
SET

FDEFINITION

PACKAGE  PACKAGEP  MAKE-PACKAGE
DEFPACKAGE
IN-PACKAGE
FIND-PACKAGE DELETE-PACKAGE
PACKAGE-NAME
PACKAGE-NICKNAMES
PACKAGE-SHADOWING-SYMBOLS
PACKAGE-USE-LIST
PACKAGE-USED-BY-LIST
RENAME-PACKAGE
WITH-PACKAGE-ITERATOR

DO-ALL-SYMBOLS LIST-ALL-PACKAGES
DO-SYMBOLS DO-EXTERNAL-SYMBOLS

  EXPORT    INTERN  SHADOW              USE-PACKAGE  IMPORT
UNEXPORT  UNINTERN  SHADOWING-IMPORT  UNUSE-PACKAGE

FIND-SYMBOL MAKE-SYMBOL COPY-SYMBOL

BIT (type)

BIT-VECTOR BIT-VECTOR-P
```

```
BIT-AND  BIT-NAND  BIT-ANDC1  BIT-ORC1  BIT-XOR  BIT-EQV
BIT-IOR  BIT-NOR   BIT-ANDC2  BIT-ORC2  BIT-NOT
BIT (function)  SBIT

KEYWORD KEYWORDP

*MODULES* PROVIDE REQUIRE

INLINE NOTINLINE
DECLARATION

ARRAY ARRAYP
ARRAY-DIMENSION ARRAY-DIMENSIONS
ARRAY-TOTAL-SIZE
ARRAY-ROW-MAJOR-INDEX ROW-MAJOR-AREF
ARRAY-DISPLACEMENT
ARRAY-ELEMENT-TYPE
ARRAY-IN-BOUNDS-P
ARRAY-RANK
ARRAY-HAS-FILL-POINTER-P
FILL-POINTER

AREF SVREF

COMPILED-FUNCTION COMPILED-FUNCTION-P
FUNCTION (type)  FUNCTIONP

FUNCTION (special form)

FUNCALL APPLY

ABORT           (function)
CONTINUE        (function)
MUFFLE-WARNING  (function)
STORE-VALUE     (function)
USE-VALUE       (function)

QUOTE IDENTITY

SETQ   PSETQ
PROGV

PROG1 PROG2 PROGN

MULTIPLE-VALUE-BIND  MULTIPLE-VALUE-CALL   MULTIPLE-VALUE-LIST  VALUES
MULTIPLE-VALUE-SETQ  MULTIPLE-VALUE-PROG1  NTH-VALUE            VALUES-LIST

SEQUENCE
MAKE-SEQUENCE  COPY-SEQ
CONCATENATE    MAP
ELT
SUBSEQ
REDUCE
MISMATCH

COUNT             DELETE          FIND
COUNT-IF          DELETE-IF       FIND-IF
COUNT-IF-NOT      DELETE-IF-NOT   FIND-IF-NOT

POSITION          REMOVE          SUBSTITUTE          NSUBSTITUTE
POSITION-IF       REMOVE-IF       SUBSTITUTE-IF       NSUBSTITUTE-IF
POSITION-IF-NOT   REMOVE-IF-NOT   SUBSTITUTE-IF-NOT   NSUBSTITUTE-IF-NOT

SUBST          NSUBST
SUBST-IF       NSUBST-IF
SUBST-IF-NOT   NSUBST-IF-NOT

SUBLIS NSUBLIS

REMOVE-DUPLICATES  DELETE-DUPLICATES


TERPRI       CLEAR-INPUT    LISTEN  FORCE-OUTPUT
FRESH-LINE   CLEAR-OUTPUT           FINISH-OUTPUT

UNREAD-CHAR
```

```
PEEK-CHAR
READ-CHAR           WRITE-CHAR
READ-CHAR-NO-HANG
LISTEN
READ-BYTE           WRITE-BYTE
READ-LINE           WRITE-LINE  WRITE-STRING
READ                PRINT       PPRINT
READ-FROM-STRING
READ-PRESERVING-WHITESPACE
                    PRINT-UNREADABLE-OBJECT
PARSE-INTEGER

WRITE             PRIN1             PRINC
WRITE-TO-STRING   PRIN1-TO-STRING   PRINC-TO-STRING

 INTERSECTION    UNION    SET-DIFFERENCE    SET-EXCLUSIVE-OR
NINTERSECTION   NUNION   NSET-DIFFERENCE   NSET-EXCLUSIVE-OR
SUBSETP

ADJOIN

OPEN CLOSE
WITH-OPEN-FILE
WITH-OPEN-STREAM
INTERACTIVE-STREAM-P
INPUT-STREAM-P
OUTPUT-STREAM-P

WITH-INPUT-FROM-STRING
WITH-OUTPUT-TO-STRING
GET-OUTPUT-STREAM-STRING

NUMBER NUMBERP
COMPLEX COMPLEXP
IMAGPART REALPART
UPGRADED-COMPLEX-PART-TYPE
REAL REALP
RATIO RATIONALP
RATIONAL RATIONALIZE
NUMERATOR DENOMINATOR
INTEGER INTEGERP
SIGNED-BYTE UNSIGNED-BYTE MOD (type)
FIXNUM BIGNUM
MOST-NEGATIVE-FIXNUM
MOST-POSITIVE-FIXNUM
FLOAT FLOATP
FLOAT-DIGITS  FLOAT-PRECISION         FLOAT-RADIX                        FLOAT-SIGN
SCALE-FLOAT   DECODE-FLOAT            INTEGER-DECODE-FLOAT
DOUBLE-FLOAT  DOUBLE-FLOAT-EPSILON    DOUBLE-FLOAT-NEGATIVE-EPSILON
LONG-FLOAT    LONG-FLOAT-EPSILON      LONG-FLOAT-NEGATIVE-EPSILON
SHORT-FLOAT   SHORT-FLOAT-EPSILON     SHORT-FLOAT-NEGATIVE-EPSILON
SINGLE-FLOAT  SINGLE-FLOAT-EPSILON    SINGLE-FLOAT-NEGATIVE-EPSILON
MOST-NEGATIVE-DOUBLE-FLOAT    MOST-POSITIVE-DOUBLE-FLOAT
MOST-NEGATIVE-LONG-FLOAT      MOST-POSITIVE-LONG-FLOAT
MOST-NEGATIVE-SHORT-FLOAT     MOST-POSITIVE-SHORT-FLOAT
MOST-NEGATIVE-SINGLE-FLOAT    MOST-POSITIVE-SINGLE-FLOAT

 FLOOR  ROUND  TRUNCATE  CEILING
FFLOOR FROUND FTRUNCATE FCEILING

LOGAND  LOGNAND LOGANDC1  LOGORC1  LOGXOR  LOGEQV
LOGIOR  LOGNOR  LOGANDC2  LOGORC2  LOGNOT  LOGTEST

INTEGER-LENGTH LOGBITP LOGCOUNT

STRING STRINGP
STRING-TRIM STRING-LEFT-TRIM STRING-RIGHT-TRIM
STRING=           STRING<           STRING>
STRING/=          STRING>=          STRING<=
STRING-EQUAL      STRING-LESSP      STRING-GREATERP
STRING-NOT-EQUAL  STRING-NOT-LESSP  STRING-NOT-GREATERP
CHAR
SCHAR

 STRING-CAPITALIZE   STRING-DOWNCASE   STRING-UPCASE
NSTRING-CAPITALIZE  NSTRING-DOWNCASE  NSTRING-UPCASE
```

```
MAKE-ARRAY MAKE-STRING
SIMPLE-VECTOR      SIMPLE-VECTOR-P
SIMPLE-BIT-VECTOR SIMPLE-BIT-VECTOR-P
SIMPLE-ARRAY
SIMPLE-STRING      SIMPLE-STRING-P
SIMPLE-BASE-STRING
BASE-STRING

LOAD-LOGICAL-PATHNAME-TRANSLATIONS
LOGICAL-PATHNAME
LOGICAL-PATHNAME-TRANSLATIONS
TRANSLATE-LOGICAL-PATHNAME
TRANSLATE-PATHNAME

CHARACTER CHARACTERP
BASE-CHAR EXTENDED-CHAR
STANDARD-CHAR
CHAR-NAME NAME-CHAR
CHAR=             CHAR<            CHAR>
CHAR/=            CHAR>=           CHAR<=
CHAR-EQUAL        CHAR-LESSP       CHAR-GREATERP
CHAR-NOT-EQUAL    CHAR-NOT-LESSP   CHAR-NOT-GREATERP
CHAR-CODE         CODE-CHAR        CHAR-INT
LOWER-CASE-P      UPPER-CASE-P     BOTH-CASE-P
CHAR-DOWNCASE     CHAR-UPCASE
ALPHA-CHAR-P
ALPHANUMERICP
DIGIT-CHAR
DIGIT-CHAR-P
GRAPHIC-CHAR-P
STANDARD-CHAR-P

MACROLET

STRUCTURE
COPY-STRUCTURE
```

The following are not quite conforming but about as close as Genera is likely to get:

```
MACROEXPAND MACROEXPAND-1  - e.g., consider (MACROEXPAND '(COND (X Y))) which
 expands into things that aren't CL-defined special forms.

DEFCONSTANT, DEFVAR, DEFPARAMETER -
   CL doesn't say these should complain if done in multiple files.

ABORT (restart) - Full integration with Zetalisp's Abort is more than we can hope for.

Condition Hierarchy:

 ARITHMETIC-ERROR
 ARITHMETIC-ERROR-OPERANDS
 ARITHMETIC-ERROR-OPERATION

 CELL-ERROR
 CELL-ERROR-NAME
 UNBOUND-SLOT
 UNBOUND-SLOT-INSTANCE
 UNBOUND-VARIABLE
 UNDEFINED-FUNCTION

 CONTROL-ERROR

 END-OF-FILE

 FILE-ERROR
 FILE-ERROR-PATHNAME

 PACKAGE-ERROR
 PACKAGE-ERROR-PACKAGE

 PRINT-NOT-READABLE
 PRINT-NOT-READABLE-OBJECT

 PROGRAM-ERROR

 READER-ERROR
```

SERIOUS-CONDITION

SIMPLE-CONDITION
SIMPLE-CONDITION-FORMAT-ARGUMENTS
SIMPLE-CONDITION-FORMAT-CONTROL

SIMPLE-ERROR

SIMPLE-TYPE-ERROR

STREAM-ERROR
STREAM-ERROR-STREAM

TYPE-ERROR
TYPE-ERROR-DATUM
TYPE-ERROR-EXPECTED-TYPE

WARNING
STYLE-WARNING

Already documented, whether conforming or not:

CLOS...

ADD-METHOD
ALLOCATE-INSTANCE
BUILT-IN-CLASS
CALL-METHOD
CALL-NEXT-METHOD
CHANGE-CLASS
CLASS
CLASS-NAME
CLASS-OF
COMPUTE-APPLICABLE-METHODS
DEFCLASS
DEFGENERIC
DEFINE-METHOD-COMBINATION
DEFMETHOD
DESCRIBE-OBJECT
DOCUMENTATION
ENSURE-GENERIC-FUNCTION
FIND-CLASS
FIND-METHOD
FUNCTION-KEYWORDS
GENERIC-FUNCTION
INITIALIZE-INSTANCE
INVALID-METHOD-ERROR
MAKE-INSTANCE
MAKE-INSTANCES-OBSOLETE
MAKE-LOAD-FORM
MAKE-LOAD-FORM-SAVING-SLOTS
MAKE-METHOD
METHOD
METHOD-COMBINATION
METHOD-COMBINATION-ERROR
METHOD-QUALIFIERS
NEXT-METHOD-P
NO-APPLICABLE-METHOD
NO-NEXT-METHOD
PRINT-OBJECT
REINITIALIZE-INSTANCE
REMOVE-METHOD
SETF
SHARED-INITIALIZE
SLOT-BOUNDP
SLOT-EXISTS-P
SLOT-MAKUNBOUND
SLOT-MISSING
SLOT-UNBOUND
SLOT-VALUE
STANDARD
STANDARD-CLASS
STANDARD-GENERIC-FUNCTION
STANDARD-METHOD
STANDARD-OBJECT
STRUCTURE-CLASS
STRUCTURE-OBJECT
SYMBOL-MACROLET
UPDATE-INSTANCE-FOR-DIFFERENT-CLASS
UPDATE-INSTANCE-FOR-REDEFINED-CLASS
WITH-ACCESSORS
WITH-SLOTS

Conditions...

BREAK-ON-SIGNALS
DEBUGGER-HOOK
ABORT
ARITHMETIC-ERROR
ARITHMETIC-ERROR-OPERANDS
ARITHMETIC-ERROR-OPERATION
ASSERT
BREAK
CCASE
CELL-ERROR
CELL-ERROR-NAME
CERROR
CHECK-TYPE
COMPUTE-RESTARTS
CONDITION
CONTINUE
CONTROL-ERROR
CTYPECASE
DEFINE-CONDITION
DIVISION-BY-ZERO
ECASE
END-OF-FILE
MAKE-CONDITION
MUFFLE-WARNING
PACKAGE-ERROR
PACKAGE-ERROR-PACKAGE
PARSE-ERROR
PRINT-NOT-READABLE
PRINT-NOT-READABLE-OBJECT
PROGRAM-ERROR
READER-ERROR
RESTART
RESTART-BIND
RESTART-CASE
RESTART-NAME
SERIOUS-CONDITION
SIGNAL
SIMPLE-CONDITION
SIMPLE-CONDITION-FORMAT-ARGUMENTS
SIMPLE-CONDITION-FORMAT-CONTROL
SIMPLE-ERROR
SIMPLE-TYPE-ERROR
SIMPLE-WARNING
STORAGE-CONDITION

ERROR	STORE-VALUE
ETYPECASE	STREAM-ERROR
FILE-ERROR	STREAM-ERROR-STREAM
FILE-ERROR-PATHNAME	STYLE-WARNING
FIND-RESTART	TYPE-ERROR
FLOATING-POINT-INEXACT	TYPE-ERROR-DATUM
FLOATING-POINT-INVALID-OPERATION	TYPE-ERROR-EXPECTED-TYPE
FLOATING-POINT-OVERFLOW	UNBOUND-SLOT
FLOATING-POINT-UNDERFLOW	UNBOUND-SLOT-INSTANCE
HANDLER-BIND	UNBOUND-VARIABLE
HANDLER-CASE	UNDEFINED-FUNCTION
IGNORE-ERRORS	USE-VALUE
INVOKE-DEBUGGER	**WARN**
INVOKE-RESTART	WARNING
INVOKE-RESTART-INTERACTIVELY	**WITH-SIMPLE-RESTART**

Not conforming as an implication of one or more other things:

```
  LAMBDA - &REST, etc.    (Do declarations scope right?)
  DEFUN  - &REST, etc.    (Do declarations scope right?)
  DEFMACRO ...
```

Non-conforming ...

[1] &REST - dynamic extent thing.

[1] *FEATURES* - Missing symbols specified in dpANS CL.

[1] READTABLE-CASE

[1] Need to decide if ANSI-Common-Lisp will be the default syntax.

[1] Need to test whether these are fully and correctly implemented and document discrepancies. If they are not, fixing them should be assigned the indicated priority.

```
[2] LET    PROG   DO                  ; binding list syntax, declaration handling
[2] LET*   PROG*  DO*                 ; binding list syntax, declaration handling
[2] EQUAL EQUALP                      ; descent rules
[2] SPECIAL declaration               ; scope
[2] IGNORE IGNORABLE                  ; scope, IGNORABLE implemented?
[2] MERGE COERCE MAP CONCATENATE      ; coercion
[2] ADJUST-ARRAY ADJUSTABLE-ARRAY-P   ; general semantics
[2] MAKE-HASH-TABLE                   ; EQUALP test, handling of rehash size/thresh
[2] IGNORE (declaration)              ; e.g., for DOTIMES, DOLIST, ...
[2] TYPE declaration                  ; particularly: its abbreviations
[2] *MACROEXPAND-HOOK*                ; SETF issues
[3] *DEFAULT-PATHNAME-DEFAULTS*       ; do the right things use this?
[3] READ-DELIMITED-LIST               ; general semantics
[4] DEFSETF                           ; arglist issues
```

[2] If EVAL-WHEN is not fully implemented, need to fix it.

[2] These restarts are probably not set up in all places where they are needed:
ABORT STORE-VALUE MUFFLE-WARNING
CONTINUE USE-VALUE

[2] OPEN-STREAM-P
FILE-STREAM — deftype

[2] DECLARE - Are all the scopes right?

[2] COMPILE-FILE, LOAD - Need :EXTERNAL-FORMAT keyword.

[2] &ALLOW-OTHER-KEYS, :ALLOW-OTHER-KEYS - recent CLOS semantics implemented?

[2] &OPTIONAL and &KEY - closures over previous vars.

[2] The type stream must be disjoint from SYMBOL.
This affects

```
STREAM STREAMP

BROADCAST-STREAM
MAKE-BROADCAST-STREAM
BROADCAST-STREAM-STREAMS

CONCATENATED-STREAM
MAKE-CONCATENATED-STREAM

ECHO-STREAM
MAKE-ECHO-STREAM
ECHO-STREAM-INPUT-STREAM
ECHO-STREAM-OUTPUT-STREAM

STRING-STREAM
MAKE-STRING-INPUT-STREAM
MAKE-STRING-OUTPUT-STREAM

SYNONYM-STREAM
MAKE-SYNONYM-STREAM
SYNONYM-STREAM-SYMBOL
```

[2] The types ECHO-STREAM and TWO-WAY-STREAM are not implemented at all.

```
ECHO-STREAM
MAKE-ECHO-STREAM
```

```
    ECHO-STREAM-INPUT-STREAM
    ECHO-STREAM-OUTPUT-STREAM

    TWO-WAY-STREAM
    MAKE-TWO-WAY-STREAM
    TWO-WAY-STREAM-INPUT-STREAM
    TWO-WAY-STREAM-OUTPUT-STREAM

    The function CONCATENATED-STREAM-STREAMS is not implemented.  [1]

[2] The new PPRINT-related format ops are not supported.
    These defined names are implicitly affected:

    Y-OR-N-P YES-OR-NO-P
    FORMAT

[2] FORMATTER is implemented, but the function it returns has the wrong return values.

[2] Other PPRINT substrate missing:

    *PRINT-LINES*
    *PRINT-MISER-WIDTH*
    *PRINT-PPRINT-DISPATCH*
    *PRINT-RIGHT-MARGIN*

    FILE-STRING-LENGTH
    COPY-PPRINT-DISPATCH
    SET-PPRINT-DISPATCH

    PPRINT-DISPATCH
    PPRINT-EXIT-IF-LIST-EXHAUSTED
    PPRINT-FILL
    PPRINT-INDENT
    PPRINT-LINEAR
    PPRINT-LOGICAL-BLOCK
    PPRINT-NEWLINE
    PPRINT-POP
    PPRINT-TAB
    PPRINT-TABULAR

[3] Contagion of mixed floats and rationals in:                    [2]
     = /=
     < <=
     > >=
     MIN, MAX

[3] STREAM-ELEMENT-TYPE returns type names in the LISP package, not the CL package
    e.g., (WITH-INPUT-FROM-STRING (STR (MAKE-STRING 3 :ELEMENT-TYPE 'BASE-CHAR))
            (STREAM-ELEMENT-TYPE STR)) → LISP:STRING-CHAR
    I believe that we should go to some care not to return things like
    LISP:STRING-CHAR from FCL substrate.  I'd prefer to see BASE-CHAR here.
    This could be done by any of the following means:
     - Special casing the specific return value for this function.
     - Having a stylized version of type-expand used generally in FCL which
       undid certain expansions for functions like this that return types.
     - Remembering the name of the type used to create the object.
       (This last might be defeated by the need to do upgrading.)

[3]  Review argument subform evaluation order
     SETF  DECF  ROTATEF  PUSH
     PSETF INCF  SHIFTF   PUSHNEW
                          POP

[3] System methods for DESCRIBE should not be doing interactive querying.
    Such querying should be provided by CP commands, perhaps by binding a special
    variable that can be tested by system methods.

[3] &KEY  (funcall (compile (defun foo ()
                              ((lambda (&key allow-other-keys) allow-other-keys)
                               :allow-other-keys nil :allow-other-keys t :foo 7))))
      fails correctly interpreted, but error not detected compiled.

[3] FUNCTION-LAMBDA-EXPRESSION should undigestify its result.
    Consider: (function-lambda-expression #'(lambda (x) x))

[3] The value of these variables, a synonym stream, should not be a symbol:
```

```
    *DEBUG-IO*          *STANDARD-INPUT*
    *ERROR-OUTPUT*      *STANDARD-OUTPUT*
    *QUERY-IO*          *TRACE-OUTPUT*

[3] Setting *READ-BASE* to >10 isn't enough to get it to read FACE as a number.

[3] Need to remove compatibility support for RETURN of multi-args.

[4] LOAD and COMPILE ignore their :PRINT argument (which implicitly means that
    *LOAD-PRINT* and *COMPILE-PRINT* don't have their desired effect).

[4] Some symbols have properties that are indicated by symbols in CL or USER or KEYWORD.

[4] The arglist of COMPILE-FILE-PATHNAME is wrong; wants &ALLOW-OTHER-KEYS.

[5] ABORT (restart)

[6] Optimization qualities not adhered to.
    OPTIMIZE
    COMPILATION-SPEED  DEBUG  SAFETY  SPACE  SPEED
```